

Name: Physics Student Roll No: 101

Q1. 6

Q2. Newtons first law says a body stays at rest or moving straight unless a force acts on it. It is the law of inertia.

Q3. Speed is distance per time, it is scalar with only magnitude. Velocity is displacement per time, it is a vector with magnitude and direction.

Q4. Energy cannot be created or destroyed only changed from one form to another. Total energy stays same. Example a falling ball changes potential energy into kinetic energy.

Q5. An electric motor changes electrical energy to mechanical energy. A coil is between magnet poles. Current in coil feels a force and rotates. The commutator reverses current each half turn so it keeps spinning one way.